Getting Started

Welcome to the A+ Grass asset! This documentation will help you get started with a quick setup of the grass in your project.

This documentation covers a brief explanation of the contents of this package, a quick setup to get started. Consider using the online documentation for in depth coverage of the A+ Grass asset as well as video demonstrations:
http://4ydam/assets/a+grass/getting-started/

Quick Setup

You can use one of the examples provided to get started quickly. To get the grass up and running in your project, simply drag and drop one of the prefabs into your scene, add Grass Manager into an empty game object, and hit play.

You can find the prefabs in the folder **A+ Grass > Prefabs > Grass Patch**

To add interactive behaviour to the grass, you need to additionally input the Grass Interactor onto any object you want interactive behaviour from. You can use the Mouse Interaction Handler script (added to your Camera) to reference an empty object with an Interactor to add mouse interactivity.

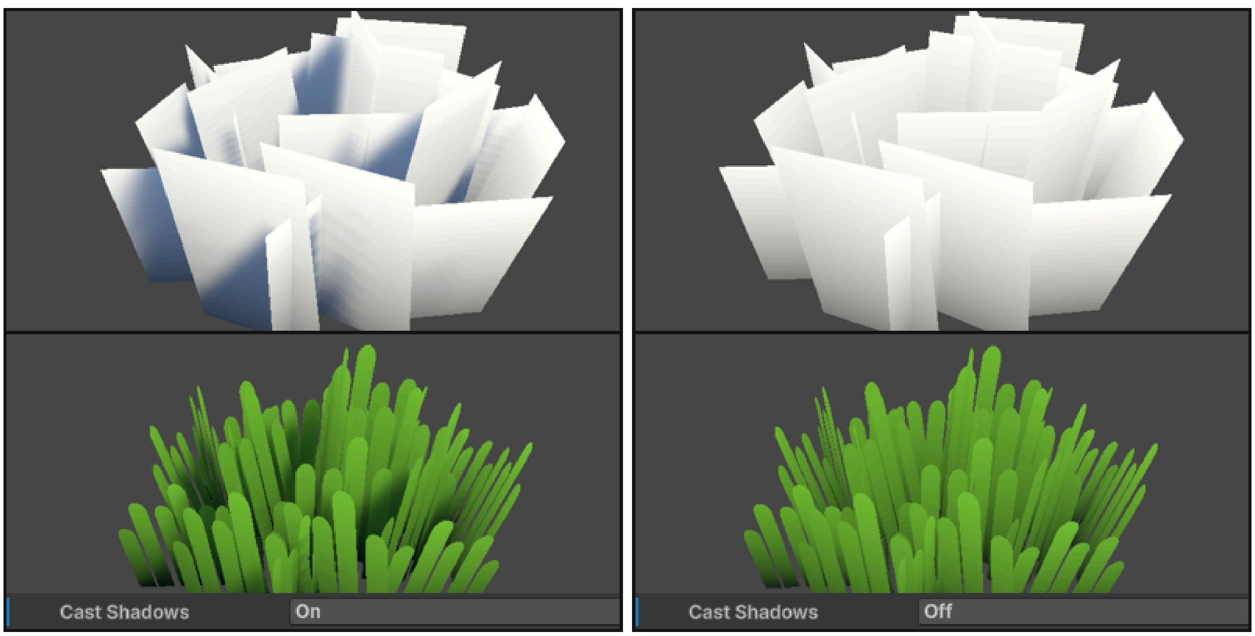
Below is a brief explanation of the asset.

Grass Patch Model

Start by adding a mesh into your scene that you can use the grass shader on. Make your own, or use the provided template located in folder **A+ Grass > Models > Grass Patch**

The template has a Level of Detail (LOD) component with 3 levels/variants of the model.

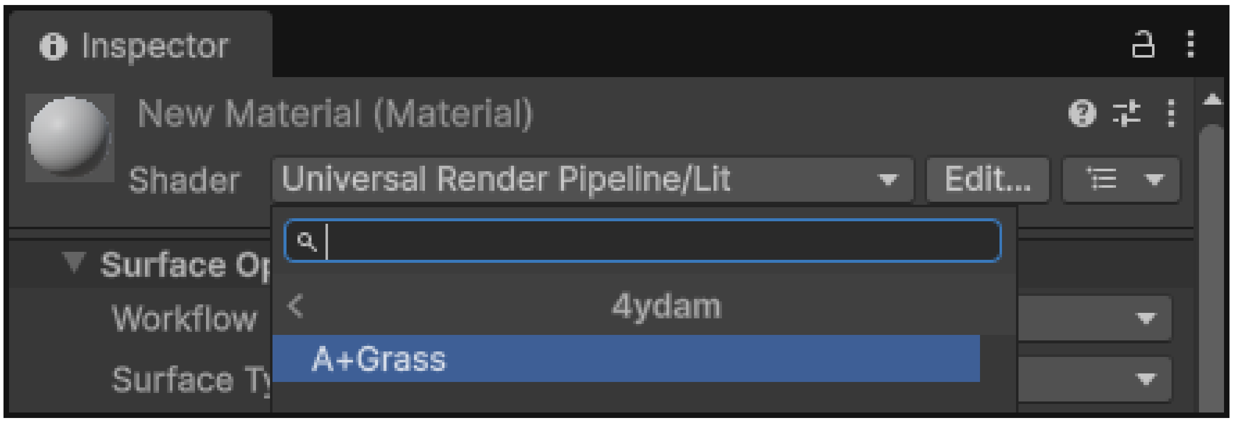
By default the Cast Shadows property is enabled in the Mesh Render, allowing the grass to cast shadows in the scene.



You should look into disabling it for each of the LOD levels, or any renderers if using a custom mesh.

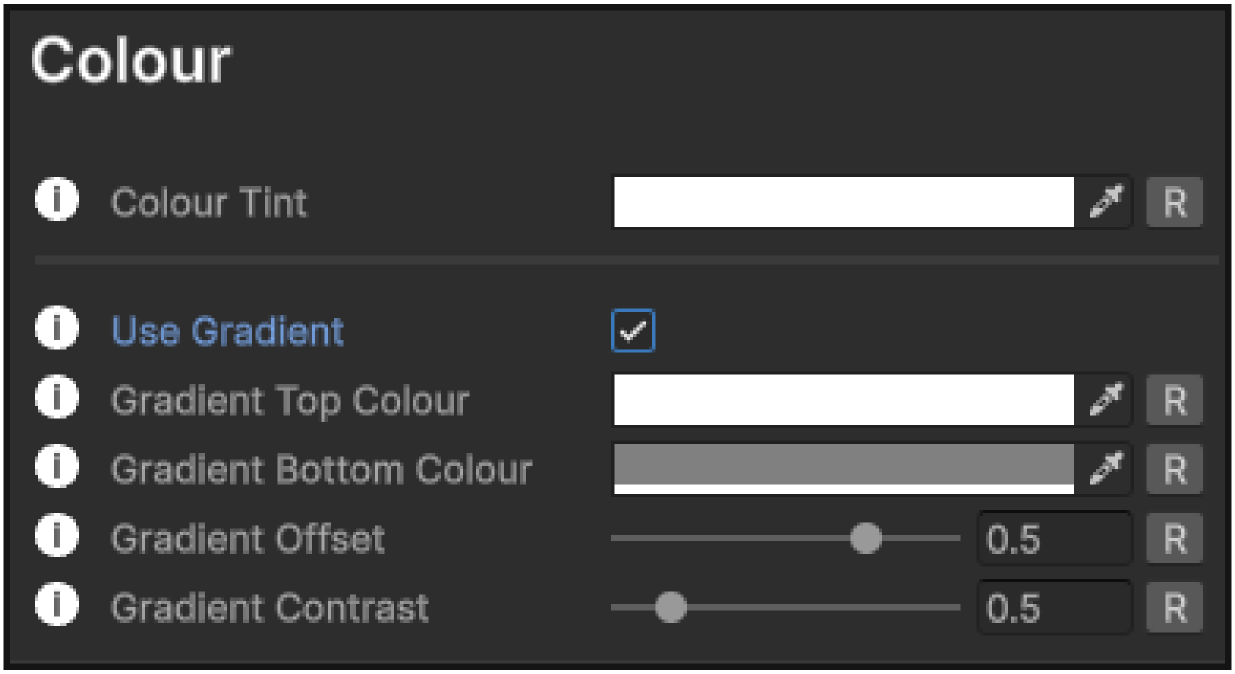
Grass Shader

You can use one of the provided Materials to get started quickly, or you can create a new material in your project and assign the shader **4ydam/A+Grass** to it. You can then assign the material to your grass mesh. Just drag and drop it onto the mesh.

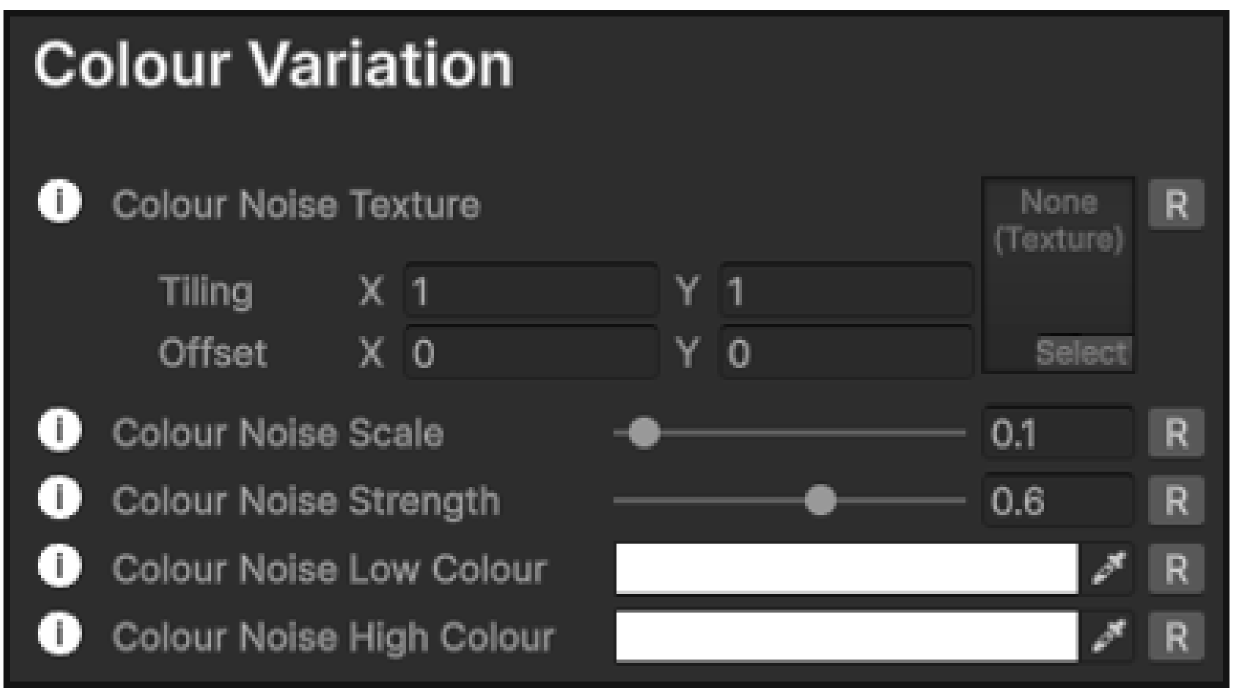


Customize the material properties to achieve the desired look for your grass.

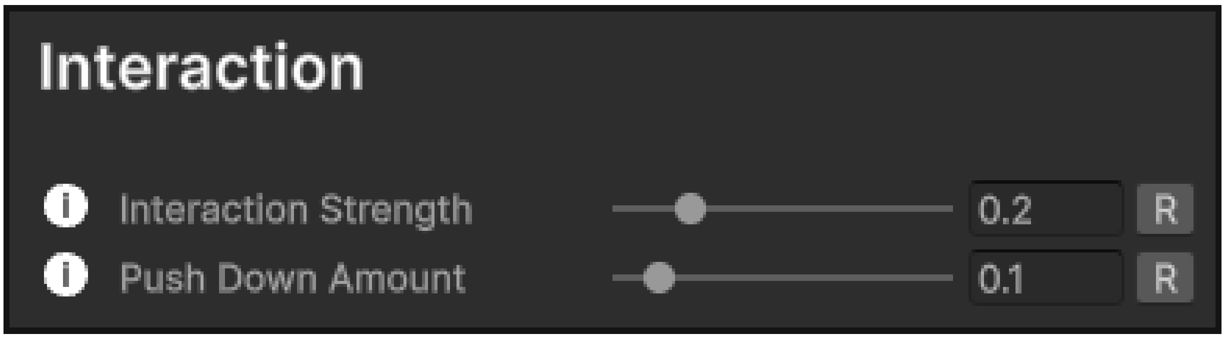
Input any 2D texture/sprite into the Texture Sample to render it as your grass. You can tile and offset it for experimental results.



There's 2 ways of setting the colour of your grass. Colour Tint (Base Colour) or if you tick Use Gradient, it will override the base colour and provide options for shaded grass. Choose between top and bottom colours and change the strength and position of the gradient using provided sliders.

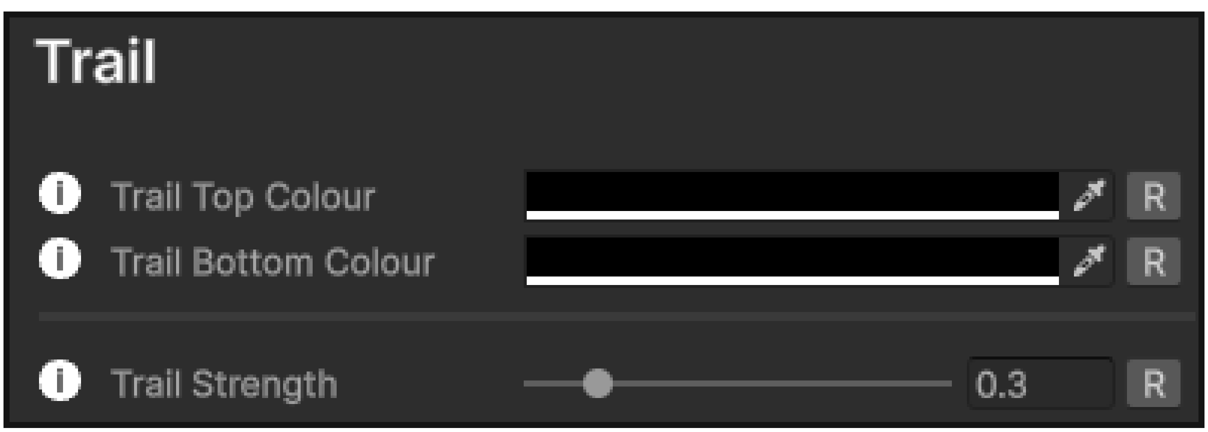


Colour Variation allows for a noise texture to be used as an overlay on top of your grass to add some depth to your grass. It uses 2 colours, applied to the white/light and black/dark areas of your noise texture, allowing for another override of the colour. Use the strength value to adjust how visible these colours are. Scale value changes the size of the noise. (Lower values are recommended for best results)

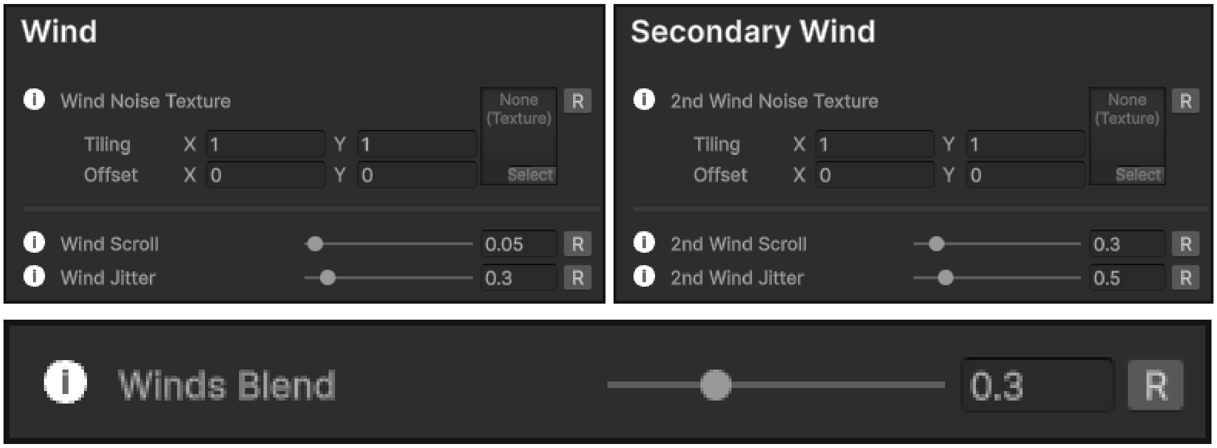


On the grass shader/material, you can adjust two parameters to control how the grass behaves when interacted with. Both should be adjusted alongside to get best results, and both values would be relatively low.

Interaction Strength controls how much the grass bends when interacted with. Push Down Amount controls how much the grass is pushed down when interacted with.

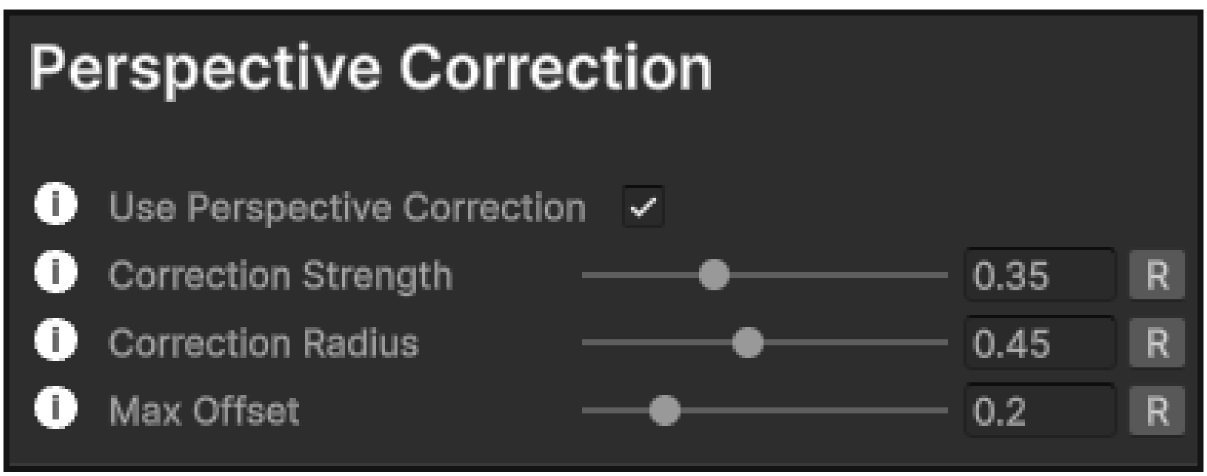


You can apply a colour on the interacted/bent grass to create a trail effect.



There are two wind noise textures that you can use to simulate a more dynamic wind effect on the grass. It works by blending the two noise textures and can be adjusted for one of them to be more dominant, or using only just one.

Wind Scroll Controls the speed of the noise texture and how strong the wind effect is. Wind Jitter Adds randomness of the wind effect, adding variation to the movement, which can make it feel more natural.

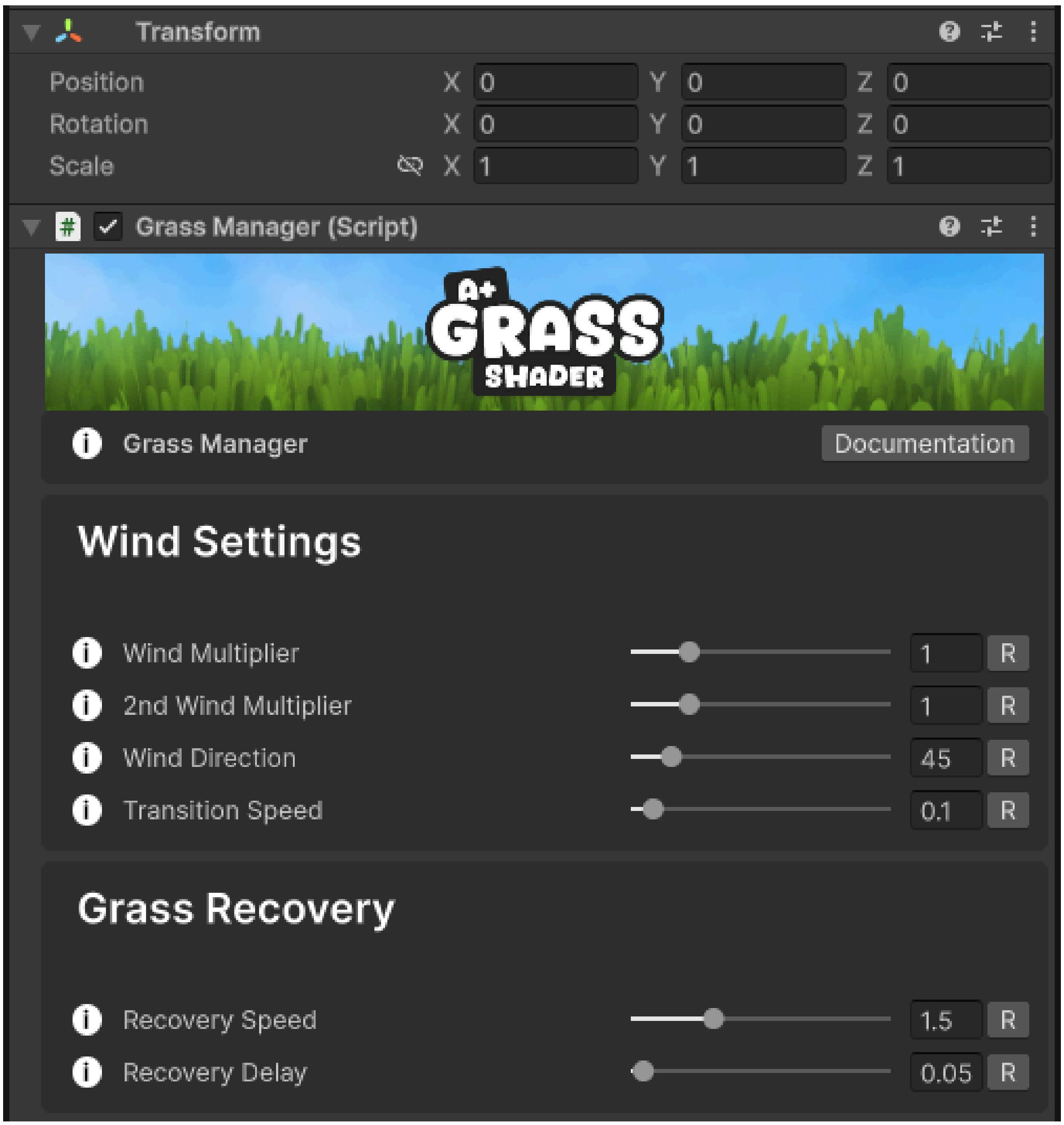


If you enable Perspective Correction, the grass will adjust its orientation based on the camera's perspective. This gives better results for a top-down game making the closer patches of grass look towards the camera. Correction Strength Controls the strength of the perspective correction effect. At the value of 0, the effect is disabled. Correction Radius Controls the starting point of the perspective correction effect when viewed from a top-down angle. Adjusting how far the effect reaches.

Max Offset Controls the maximum offset applied by the perspective correction effect, limiting the strength of the correction.

Grass Manager

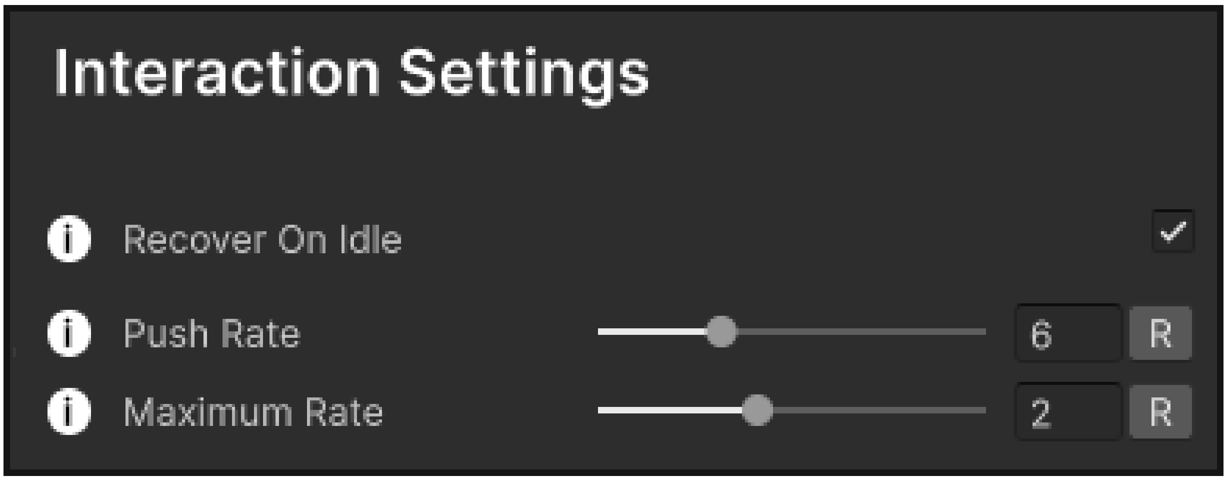
The Grass Manager is responsible for managing all interactive grass elements in the scene. There are also global settings for the wind you can adjust at runtime, and recovery of the grass from interactions to change how quickly it recovers.



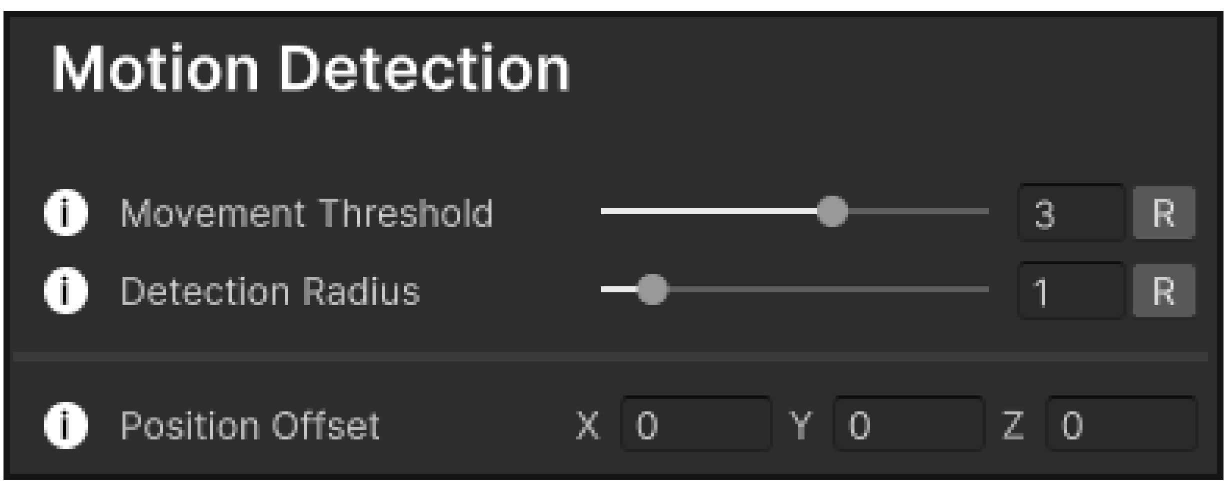
Add it to an empty game object in your scene and adjust accordingly.

Interactor

The Grass Interactor is responsible for handling the interactions between the grass and other objects in the scene. It allows objects to affect the grass, bending and pushing it. You can have as many interactors in the scene, just make sure to add the Grass Interactor component to each.



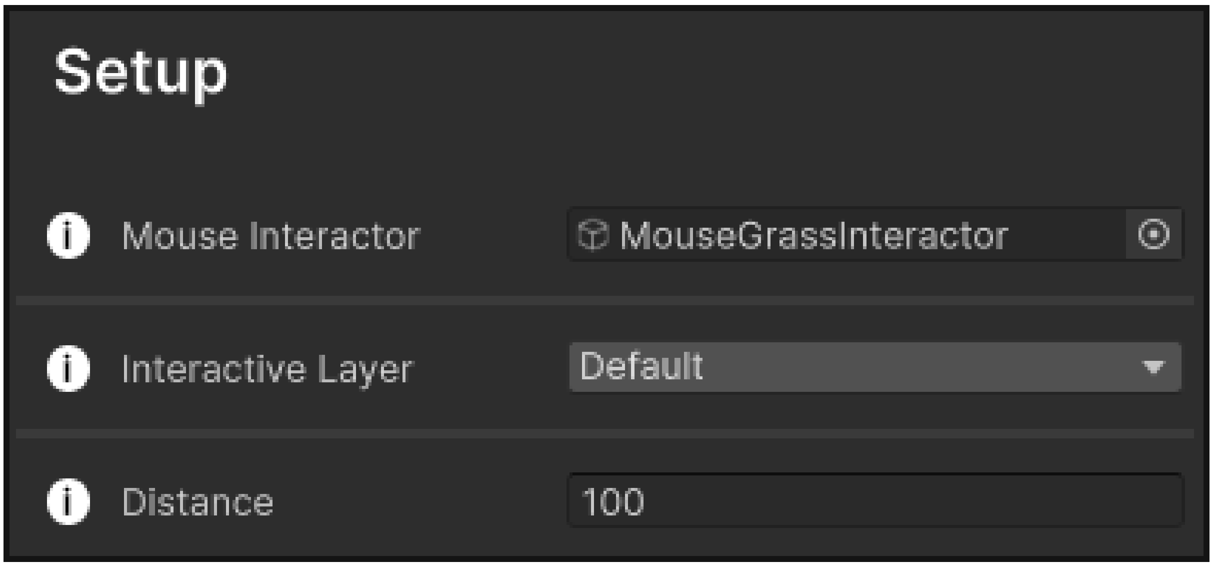
Adjust how much the grass gets pushed down. Push Rate accumulates with each interaction, limited by Maximum Rate.



Threshold limits at what speed interaction takes place. Detection Radius changes the size of the interactor. Position Offset lets you change the center point of the interactor.

Mouse Interaction Handler

The Mouse Grass Interaction Handler provides data from the mouse position off the camera using a ray to allow an interactor to follow the mouse position and interact with the grass.



Assign an empty game object with the Interactor component to the Mouse Grass Interaction Handler. This gameObject will follow the mouse position and interact with the grass where the ray hits.

For further information, go to: http://4ydam/assets/a+grass/getting-started/

For any questions/inquiries about the asset, contact me at: support@4ydam.com

If you like the asset, consider leaving a review on the store page! Cheers, 4ydam